

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

Course Name:	COMPUTER VISION	Course Code:	ITA05
Student Name:	SRIRAM R	Ref. No:	192525198
Outcome:	CO2: Apply image enhancement and segmentation techniques (BTL 3)		

Question 1: Grey Level Transformation

(a) Purpose of Grey Level Transformation in Image Enhancement

Grey level transformations map input pixel intensity values r to output pixel intensity values s using a functional transformation $s = T(r)$. The main objectives include:

- **Contrast Enhancement:** Dynamic expansion of pixel intensity ranges to improve overall image visibility and feature contrast.
- **Noise Reduction & Pre-processing:** Adjusts signal levels for standardized downstream computer vision algorithms.
- **Feature Highlighting:** Isolates specific intensity bands while dimming irrelevant visual backgrounds.

(b) Application of Negative Transformation

Given the negative transformation equation $s = 255 - r$ and input values $r = [50, 100, 150, 200]$:

- For $r = 50$: $s = 255 - 50 = 205$
- For $r = 100$: $s = 255 - 100 = 155$
- For $r = 150$: $s = 255 - 150 = 105$
- For $r = 200$: $s = 255 - 200 = 55$

Transformed Pixel Values: $s = [205, 155, 105, 55]$

Question 2: Log Transformation

(a) How Log Transformation Enhances Low-Intensity Pixels

The standard logarithmic transformation formula is $s = c \cdot \log(1 + r)$. Since the logarithmic curve grows steeply near the origin ($r \rightarrow 0$) and flattens out at higher intensity values:

- It spreads out narrow ranges of dark, low-intensity pixel values into wider output intensity bands.
- It compresses high-intensity (bright) pixel regions.
- This mechanism reveals hidden structural details within dark image regions (e.g., Fourier spectrum images).

(b) Calculation of Log Transformation

Given $s = c \cdot \log_{10}(1 + r)$ with $c = 10$ and inputs $r = [0, 10, 50, 100]$:

- For $r = 0$: $s = 10 \cdot \log_{10}(1 + 0) = 10 \cdot 0 = 0$
- For $r = 10$: $s = 10 \cdot \log_{10}(11) \approx 10 \cdot 1.0414 = 10.41$
- For $r = 50$: $s = 10 \cdot \log_{10}(51) \approx 10 \cdot 1.7076 = 17.08$
- For $r = 100$: $s = 10 \cdot \log_{10}(101) \approx 10 \cdot 2.0043 = 20.04$

Transformed Pixel Values: $s = [0, 10.41, 17.08, 20.04]$

Question 3: Histogram Processing

(a) Role of Histogram Equalization in Contrast Enhancement

Histogram equalization redistributes intensity levels so that the output image histogram approaches a uniform distribution across the dynamic range. It flattens the Cumulative Distribution Function (CDF) to maximize information entropy and improves global contrast, making low-contrast features clearly discernible.

(b) Computation of Cumulative Distribution and Equalized Values

For a 3-bit image, the intensity levels are $L = 2^3 = 8$ (ranging from 0 to 7). Total pixels $N = \sum n_i = 2 + 2 + 4 + 8 = 16$.

Equalized intensity formula: $s_k = \text{round}((L - 1) / N \cdot \text{CDF}(k)) = \text{round}((7 / 16) \cdot \text{CDF}(k))$

Intensity (k)	Frequency (n_k)	Cumulative Frequency (CDF)	Calculation ($7/16 \times \text{CDF}$)	Equalized Intensity (s_k)
0	2	2	0.875	1
1	2	4	1.750	2
2	4	8	3.500	4
3	8	16	7.000	7

Equalized Intensity Mapping: $[0 \rightarrow 1, 1 \rightarrow 2, 2 \rightarrow 4, 3 \rightarrow 7]$

Question 4: Spatial Filtering (Smoothing)

(a) Purpose of Spatial Filtering for Smoothing

Spatial smoothing filters (low-pass spatial filters) attenuate high-frequency variations in intensity. They are primary tools for noise reduction (e.g., removing spurious camera noise) and pre-processing tasks like blurring fine details prior to large-scale feature extraction or segmentation.

(b) 3×3 Averaging Filter Calculation

Given 3×3 image region:

10	20	30
20	30	40
30	40	50

$$\text{Sum} = 10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

$$\text{New Center Pixel Value} = \text{Sum} / 9 = 270 / 9 = 30$$

New Center Pixel Value: 30

Question 5: Spatial Filtering (Sharpening)

(a) How Sharpening Enhances Image Details

Sharpening highlights fine details, edges, and discontinuities by employing high-pass spatial filters based on derivative operators (such as the Laplacian). Derivative operators produce significant response at spatial transitions while yielding zero in constant background regions, thereby enhancing edge sharpness when combined with the original image.

(b) Application of Laplacian Mask

Laplacian Mask M and Image Region I :

Mask (M):

0	-1	0
-1	4	-1
0	-1	0

Image Region (I):

10	10	10
10	20	10
10	10	10

$$\text{Output} = (0 \times 10) + (-1 \times 10) + (0 \times 10) + (-1 \times 10) + (4 \times 20) + (-1 \times 10) + (0 \times 10) + (-1 \times 10) + (0 \times 10)$$

$$\text{Output} = 0 - 10 + 0 - 10 + 80 - 10 + 0 - 10 + 0 = 40$$

Output Center Pixel Value: 40